MUHAMMAD ZEESHAN SHAH,

Ph.D. Candidate – Defending Oct 19, 2026

Biomaterials & Polymer Chemist | Liposomal & Lipid Nanoparticle Formulation | Bioconjugation & Immunoengineering Platforms

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EXECUTIVE SUMMARY

Results-driven **Biomaterials and Polymer Scientist** with **7+ years of total research experience** (5+ years of doctoral research at Miami University plus 2 years of M.S. research) engineering **liposomal, lipid-based, and polymeric nanoparticle systems**, **bioconjugated copolymers**, and **antiviral/biomimetic lipid assemblies**. Proven scientific leadership across **9 peer-reviewed publications** (*Biomacromolecules*, *ACS Macro Letters*, *Polymer Chemistry*, *Polymer*, *BBA-Biomembranes*) and **6 manuscripts under review/in preparation**. Expert in **controlled RAFT polymerization**, **bioconjugation and surface functionalization (PEGylation, amino acid conjugation, and charge-tunable modification)**, **liposome/vesicle preparation and nanoparticle formulation (**$3–60nm$**)**, and **multi-scale analytical characterization (HPLC, SEC/GPC, DLS, TEM/SEM, NMR, CW-EPR, Rheometry)**. Experienced laboratory leader with a track record of **mentoring 13 junior researchers (11 undergraduates, 2 graduate students)**, coordinating multi-lab **NIH-funded projects**, and managing laboratory operations, inventory, and SOPs.

EDUCATION

- **Ph.D. in Chemistry (Polymer Science, Biomaterials & Lipid Nanoparticle Engineering)** | Miami University, Oxford, OH | **Oct 2026**
    - *Dissertation: "Vinyl Ether-Maleic Anhydride Copolymers Facilitating Native-Like Lipid Membrane Environments"* (Defense: Oct 19, 2026)
    - *Advisors:* Prof. Dominik Konkolewicz & Prof. Rick Page (in collaboration with Prof. Gary Lorigan)
- **M.S. in Material Chemistry** | Miami University, Oxford, OH | **2023**
- **M.S. in Physical Chemistry** | Government College University, Faisalabad | **2017**

CORE COMPETENCIES

- **Liposomal, Lipid-Based & Polymeric Biomaterials:** Liposome & Phospholipid Vesicle Preparation (POPC, POPG, DMPC), Lipid Nanoparticles (SMALPs, VEMALPs), Polymeric Nanoparticles & Micelles, Antiviral Bilayer-Disrupting Polymer Networks, Rapid Aqueous Self-Assembly ($< 60s$), Scalable Thin-Film Nanoparticle Fabrication.
- **Chemical Synthesis, Bioconjugation & Surface Functionalization:** Reversible Addition-Fragmentation Chain Transfer (RAFT) Polymerization, Small-Molecule Chain Transfer Agent (CTA) Synthesis, Block & Alternating Copolymerization, Nucleophilic Ring-Opening Bioconjugation (PEGylation, Amino Acid Conjugation, Cationic/Anionic/Zwitterionic Surface Modification), Cross-Linked Hydrogel & Sorbent Networks.
- **Nanoparticle Characterization & Analytical Toolkit:** Dynamic Light Scattering (DLS), Transmission & Scanning Electron Microscopy (TEM, SEM), High-Performance Liquid Chromatography (HPLC), Size Exclusion / Gel Permeation Chromatography (SEC/GPC), 1D ^{1}H and DOSY NMR, Continuous-Wave

Electron Paramagnetic Resonance (CW-EPR), UV-Vis, FTIR, Rheometry, Instron Mechanical Testing, DSC, ICP-OES, AAS.

- **Biological Interface & Reconstitution Assays:** Native-Like Reconstitution & Stabilization of Collaborator-Purified Membrane Proteins (**Human TRPV1**, **KCNE1**, **KCNE4**, **gp28**) in Proteoliposomes and Nanodiscs, Physiological & Ionic Stability Profiling (pH 3.0–9.0, up to 30 mM Mg^{2+}), Vesicle Remodeling Assays.
- **Laboratory Leadership, Mentorship & Operations:** Multi-Investigator NIH Project Coordination, Undergraduate & Graduate Mentorship (13 trainees), Laboratory Inventory & Procurement Management, Equipment Maintenance & SOP Development, Grant Progress Reporting & Manuscript Preparation.
- **Computational & Design Tools:** KaleidaGraph, Bruker TopSpin, Fiji/ImageJ, Autodesk Fusion 360 (CAD & 3D Printing Fabrication), Adobe Illustrator, Python (Kinetic Modeling & Data Visualization basics), MATLAB-basics.

RESEARCH EXPERIENCE

Graduate Research Fellow – Biomaterials, Lipid Nanoparticles & Chemical Synthesis

Miami University | *Laboratories of Prof. Dominik Konkolewicz, Prof. Gary Lorigan & Prof. Rick Page (NIH- & NSF-Funded)* | **May 2021 – Present**

- **Liposomal, Lipid-Based & Polymeric Nanoparticle Engineering:** Formulated and characterized biomimetic phospholipid liposomes (POPC, POPG, DMPC) and self-assembled polymer–lipid nanoparticles (VEMALPs and SMALPs). Engineered chain-extended amphiphilic block copolymers (VEMA-b-nBA and VEMA-b-tBA/AA) that precisely tune nanoparticle diameters from **60 nm down to 3 nm** while preserving payload integrity (*Polymer Chemistry* 2026; *BBA-Biomembranes* 2025; *Biophysical Chemistry* 2025).
- **Bioconjugation & Surface Functionalization (PEG, Amino Acids & Charge Tuning):** Exploited reactive maleic anhydride moieties along the copolymer backbone for modular nucleophilic bioconjugation with **polyethylene glycol (PEG)** chains, **amino acids**, and cationic, anionic, and zwitterionic modifiers to yield **9 distinct charge-tunable derivatives (VEMADs)** (*Polymer Chemistry* submitted; manuscripts in preparation). Demonstrated complete resistance to aggregation across **pH 3.0–9.0** and up to **30 mM** Mg^{2+}.
- **Precision RAFT Chemical Synthesis & Small-Molecule CTA Screening:** Spearheaded the rational synthesis and kinetic modeling of alternating and block **vinyl ether–maleic anhydride (VEMA)** copolymers via RAFT polymerization; synthesized and screened 4 small-molecule CTAs (iBADTC, PADTC, EtPAX, PAETC) to achieve narrow dispersity ($Đ \sim 1.3–1.5$) and programmable molecular weights ($M_n = 6–40 kDa$) (*Biomacromolecules* 2024; *Polymer* 2026).
- **Rapid (<60 s) Mild Nanoparticle Formulation for Sensitive Biologics:** Designed aliphatic **cyclohexyl-VEMA (CH-VEMA)** copolymers ($6kDa$ and $12kDa$) that achieve **spontaneous aqueous self-assembly with lipid bilayers in under 60 seconds** without mechanical shear, organic co-solvents, or freeze–thaw cycles, maintaining monodisperse $\sim 10nm$ colloidal stability for over 72 hours. Co-developed scalable thin-film fabrication methods for lipid nanoparticles (*Analytical Biochemistry* 2022).
- **Antiviral Polymer Networks & Biomimetic Vesicle Remodeling:** Co-engineered cross-linked polymer networks incorporating lipid-bilayer disrupting macromonomers for **antiviral applications** (*Polymer Chemistry* 2022), reusable chelation polymer networks (*ACS Macro Letters* 2026), and carbon-dot-induced remodeling of biomimetic POPC vesicles (*ACS Omega* submitted).

- **Proteoliposome Reconstitution & Multi-Scale Biophysical Characterization:** Encapsulated and stabilized collaborator-purified membrane proteins (**human TRPV1**, **KCNE1**, **KCNE4**, and **gp28**) within POPC/POPG proteoliposomes and polymer nanodiscs. Integrated **HPLC, SEC/GPC, DLS, TEM, SEM, 1D ^1H/DOSY NMR, Rheometry, and Instron** with site-directed spin labeling **CW-EPR spectroscopy** to evaluate nanoparticle morphology, membrane fluidity, and protein conformational stability (*Chemistry and Physics of Lipids* 2025).
- **Laboratory Operations, Project Coordination & Mentorship:** Coordinated interdisciplinary experimental milestones across synthetic polymer and membrane biophysics labs for NIH-funded projects; managed chemical ordering, inventory, instrument maintenance, and safety/experimental SOPs while directly mentoring **11 undergraduate and 2 graduate researchers**.

Graduate Research Assistant – Polymer Chemistry & Formulation Science

Government College University Faisalabad | *Laboratory of Dr. Munir* | **2015 – 2017**

- Investigated the radiolytic degradation kinetics of organic compounds using **Gamma Irradiation** and evaluated amphiphilic copolymer formulations for aqueous contaminant sequestration.

PEER-REVIEWED PUBLICATIONS

Published & In-Press Journal Articles

1. **Shah, M.Z.**, Rotich, N.C., Okorafor, E.A., Oestreicher, Z., Demidovich, G., Eapen, J., Page, R.C., Konkolewicz, D., & Lorigan, G.A. (2024). Vinyl ether maleic acid polymers: tunable polymers for self-assembled lipid nanodiscs and environments for membrane proteins. ***Biomacromolecules***, 25(10), 6611–6622.
2. **Shah, M.Z.**, Okorafor, E., Rotich, N.C., Henoch, Q., Thapa Acharya, R., Page, R.C., Lorigan, G.A., & Konkolewicz, D. (2026). Vinyl ether maleic acid block copolymers: a versatile platform for tunable self-assembled lipid nanodiscs and membrane protein characterization. ***Polymer Chemistry***, 17(2), 194–206.
3. Acharya, R.T., **Shah, M.Z.**, Woodruff, A., & Konkolewicz, D. (2026). Vinyl Ether Maleic Anhydride Copolymers: Efficient and Reusable Sorbents for Removing Heavy Metals from Water. ***ACS Macro Letters***, 15(7), 1035–1041.
4. Acharya, R.T., **Shah, M.Z.**, Woodruff, A., & Konkolewicz, D. (2026). Kinetics of RAFT Copolymerization of Vinyl Ether and Maleic Anhydride Monomers. ***Polymer***, 129999.
5. Okorafor, E.A., Gordon, E.A., Sahu, I.D., **Shah, M.Z.**, Konkolewicz, D., & Lorigan, G.A. (2025). Influence of lipid saturation on the structural properties of styrene maleic acid lipid nanoparticles (SMALPs). ***Biochimica et Biophysica Acta (BBA) - Biomembranes***, 1867(5-6), 184424.
6. Gordon, E.A., Okorafor, E.A., Sahu, I.D., Burridge, K.M., **Shah, M.Z.**, Thapa, O., Konkolewicz, D., & Lorigan, G.A. (2025). The effect of lipid saturation on the formation of styrene maleic acid lipid nanoparticles. ***Biophysical Chemistry***, 107566.
7. Rotich, N.C., Okorafor, E.A., Sahu, I.D., **Shah, M.Z.**, Konkolewicz, D., & Lorigan, G.A. (2025). A comparison of the effect of SMA derivatives on the structural topology and dynamics of two bacteriophage peptides. ***Chemistry and Physics of Lipids***, 105562.
8. Gordon, E.A., Richardson, Y.B., **Shah, M.Z.**, Burridge, K.M., Konkolewicz, D., & Lorigan, G.A. (2022). Formation of styrene maleic acid lipid nanoparticles (SMALPs) using SMA thin film on a substrate. ***Analytical Biochemistry***, 647, 114692.

9. Burridge, K.M., Rahman, M.S., De Alwis Watuthanthrige, N., Gordon, E., **Shah, M.Z.**, et al. (2022). Network polymers incorporating lipid-bilayer disrupting polymers: towards antiviral functionality. ***Polymer Chemistry***, 13(31), 4547–4556.

Manuscripts Submitted & In Preparation

10. **Shah, M.Z.**, et al. (2026). Charge-Tunable Vinyl Ether-Maleic Acid Derivatives (VEMADs) for Membrane Protein Solubilization and Stability. ***Polymer Chemistry*** *(Submitted)*.
11. Nusair, N.A., Bhowmick, M., Lorigan, G.A., **Shah, M.Z.**, Renaud, J.F., Schlabach, A.D., van Straaten, L.M., Davies, E.S., Rotich, N.C., & Huff, E.R. (2026). Ascorbic Acid-Derived Carbon Dots Induce Concentration-Dependent Remodeling of Biomimetic POPC Vesicles. ***ACS Omega*** *(Submitted)*.
12. **Shah, M.Z.** & Konkolewicz, D. (2026). Cyclohexyl Vinyl Ether-Maleic Acid (CH-VEMA) Copolymers for Rapid Solubilization of Lipid Bilayers. *(Manuscript in preparation)*.
13. Cubberly, B., **Shah, M.Z.**, & Konkolewicz, D. (2026). Derivation of Vinyl Ether Maleic Anhydride Copolymers using Polyethylene Glycol (PEG) Chains for Membrane Protein Study. *(Manuscript in preparation)*.
14. Hicov, N., **Shah, M.Z.**, & Konkolewicz, D. (2026). Derivation of Vinyl Ether Maleic Anhydride Copolymers using Amino Acids for Membrane Protein Study. *(Manuscript in preparation)*.
15. Okorafor, E.A., Sahu, I.D., **Shah, M.Z.**, Magar, N., Ogundepo, D.M., Perera, P., Hicov, N., Morris, A.K., McCarrick, R.M., Dabney-Smith, C., Konkolewicz, D., & Lorigan, G.A. (2026). VEMA as a Promising Alternative to SMA for Polymer Lipid Nanodiscs Biophysical Structural Studies. *(Manuscript in preparation)*.

SCIENTIFIC LEADERSHIP, MENTORSHIP & LABORATORY MANAGEMENT

- **Lead Research Mentor (11 Undergraduates & 2 Graduate Students)** | Miami University (**2022 – Present**)
    - Directly trained and supervised 13 researchers in RAFT synthesis, PEG/amino-acid bioconjugation, liposome/nanoparticle formulation, NMR/GPC/DLS analysis, and scientific communication, resulting in **5+ co-authored manuscripts** (including Cubberly et al. and Hicov et al., 2026).
- **President** | American Chemical Society (ACS) POLY/PMSE Cincinnati Chapter (**2024 – 2025**)
    - Spearheaded cross-institutional symposia, poster judging, and networking initiatives connecting academic polymer and biomaterials scientists with industrial leaders; previously served as **Treasurer (2022 – 2023)**.
- **Graduate Teaching Assistant** | Department of Chemistry & Biochemistry, Miami University (**2021 – 2023**)
    - Instructed laboratory and recitation sections for **Organic Chemistry I & II** (CHM 244/245) and **General Chemistry I & II** (CHM 144/145), overseeing laboratory safety, chemical inventory, and experimental grading.
- **Assistant Education Officer** | School Education Department, Punjab (**2017 – 2020**)
    - Directed multi-institutional operations, budget/inventory auditing, personnel evaluation, and STEM curriculum implementation across public and private schools.
- **Peer Reviewer** | *Journal on Excellence in College Teaching* (**2023 – Present**)

HONORS, FELLOWSHIPS & AWARDS

- **Poster Judge & PMSE Division Representative** | ACS Fall 2025 National Meeting, Washington, DC (Aug 2025)
- **Best Presenter Award** | American Chemical Society (ACS) Southwest Ohio Chapter (Oct 2024)
- **Graduate Research Fellowship (2x Recipient)** | Miami University (May 2022 & May 2023)
- **Competitive Graduate Travel Awards (3x Recipient)** | ACS Fall 2025 (Washington, DC), SMALPs Conference 2023 (Knoxville, TN), and ACS Spring 2023 (Indianapolis, IN)

SELECTED NATIONAL & INTERNATIONAL PRESENTATIONS

1. **Shah, M.Z.** & Konkolewicz, D. *Vinyl Ethers–Maleic Anhydride Lipid Nanodiscs: Synthesis, Block Extension, and Biophysical Performance.* (Dual Poster Presentations, **PMSE & POLY Divisions**), **ACS Fall 2025 National Meeting**, Washington, DC (Aug 2025).
2. **Shah, M.Z.** *Polymer Systems for Studying Membrane Proteins in Native-Like Environments.* (Invited Oral Presentation), **International Conference on Functional Materials and Emerging Technologies**, Faisalabad (Nov 2024).
3. **Shah, M.Z.** *Polymer-Based Lipid Nanodiscs: Tunable Macromolecular Design.* (Oral Presentation), **Graduate Research Forum**, Miami University, OH (Nov 2024).
4. **Shah, M.Z.** *Vinyl Ethers–Maleic Anhydride Lipid Nanodiscs.* (Award-Winning Oral Presentation), **ACS Southwest Ohio Chapter Symposium** (Oct 2024).
5. **Shah, M.Z.** *Tunable VEMA Copolymers for Membrane Protein Extraction.* (Oral & Poster Presentations), **SMALPs Conference**, University of Tennessee, Knoxville, TN (Nov 2023).
6. **Shah, M.Z.** *Macromolecular Engineering of Amphiphilic Copolymers.* (Flash Talk), **ACS POLY/PMSE Symposium**, University of Cincinnati, OH (July 2023).
7. **Shah, M.Z.** *Kinetic Study of Butyl Vinyl Ether and Maleic Anhydride Copolymerization & Lipid Disc Formation.* (2 Oral Presentations), **ACS Spring 2023 National Meeting**, Indianapolis, IN (March 2023).